A strict saddle basin for Schedule-Free gradient descent

Abstract

We construct a fixed C^∞ coercive function on $\mathbb{R}^2$, with globally 1-Lipschitz gradient and exactly three nondegenerate critical points, for which standard unperturbed Schedule-Free gradient descent converges to a strict saddle from an open set of position initializations. The momentum parameter is $9/10$, the step size is $1/2$, and the averaging coefficient is $1/(k+1)$. Every phase map after the first update is a global diffeomorphism, and the initialized surface retains rank two at every finite time. The construction places an interval of initialized states on an exact decaying solution of the local negative-quadratic recurrence. A finite affine interpolation and an even mollifier realize all required queries in one globally smooth objective. An explicit energy controls the positive-quadratic coordinate. The example realizes the initialization obstruction in existing perturbed saddle-avoidance analyses; it concerns the initialized surface and does not assert attraction from an open set of ambient phase states.

Keywords: Discrete dynamical system; nonautonomous iteration; strict saddle; initialization

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1 Introduction

Consider the deterministic Schedule-Free iteration

$$\begin{aligned} y_k &= (1-\beta)z_k + \beta x_k, \\ z_{k+1} &= z_k - \gamma \nabla f(y_k), \\ x_{k+1} &= \frac{k x_k + z_{k+1}}{k+1}, \qquad k \geq 0, \end{aligned} \tag{1}$$

with the standard initialization $x_0 = z_0 = a$. We fix

$$\beta = \frac{9}{10}, \qquad \gamma = \frac{1}{2}.$$

In particular, $x_1 = z_1$. The point y_k is the gradient query, whereas x_k is the usual evaluation sequence. Their distinction will be important in the construction.

Chae and Kim [1] study (1) as a nonautonomous dynamical system. Their Theorem 5.2 proves strict-saddle avoidance after an independent absolutely continuous perturbation of z_1. The paper identifies why ambient measure-zero stable-set arguments do not directly give the same conclusion for standard initialization: the state at time one lies on a diagonal of lower dimension. A set of ambient measure zero may have positive induced measure on that diagonal. We give an explicit example in which this possibility actually occurs.

The distinction between phase-space initialization and equal-start initialization has earlier precedents. O'Neill and Wright [2] analyze the local diagonal and center-stable subspace for accelerated methods while separating the global equal-start question. Dixit, Gürbüzbalaban and Bajwa [3] also give equal-start results within their accelerated-gradient framework. Their displayed scheme uses one common extrapolate for both the nongradient update and the gradient query, a fixed step, and finite limiting extrapolation parameters. Eliminating z_k in (1), however, gives for $k \geq 1$

$$\begin{aligned} x_{k+1} = x_k + \frac{k-1}{k+1}(x_k - x_{k-1}) \\ - \frac{\gamma}{k+1}\nabla f\big(x_k + (1-\beta)(k-1)(x_k - x_{k-1})\big). \end{aligned} \tag{2}$$

The query and update extrapolations are different, and the query coefficient is unbounded. Thus those results do not apply by identifying the two recurrences. The nonautonomous center-stable set theorem of Muşat and Boumal [4] supplies ambient avoidance results; restricting them to a noninvariant initialized surface requires additional information.

Positive-measure saddle basins themselves are not new. Schaeffer and McCalla [5, Example 2.3] construct smooth gradient-descent examples that focus a region into a stable manifold through a degenerate update. A more recent example of Muşat and Boumal [6, Remark 6] maps an exterior region to a saddle in one step. These are relevant precedents for smooth global constructions. In the example below, the step satisfies $\gamma L < 1$, the initialized position map is invertible, and all later phase maps are diffeomorphisms. The mechanism is alignment of the initialized surface with a stable set, with its own dimension preserved.

Our principal result is the following.

Theorem 1. *There exists an explicitly specified C^∞ coercive function $f : \mathbb{R}^2 \to \mathbb{R}$ with globally 1-Lipschitz gradient and exactly three nondegenerate critical points, such that the following assertions hold for* (1).

1. *The origin is a strict saddle, and $f(u, v) = (u^2 - v^2)/4$ in a neighborhood of the origin. The other two critical points are strict local minima.*

2. *For every standard initial point*

$$a \in \mathcal{U} = (-1, 1) \times (1 - 10^{-8}, 1 + 10^{-8}), \tag{3}$$

   one has $x_k \to 0$, $z_k \to 0$ and $y_k \to 0$. Thus the initialized saddle basin has planar Lebesgue measure at least $4 \cdot 10^{-8}$.

3. *Every full phase map $(x_k, z_k) \mapsto (x_{k+1}, z_{k+1})$ for $k \geq 1$ is a global C^∞ diffeomorphism of $\mathbb{R}^4$. The time-zero map on the initialized diagonal has an invertible position map. The initialized surface therefore has rank two at every finite time.*

The function and all its parameters are fixed independently of a. The negative-coordinate state remains nonzero at every time $k \geq 10$.

For example, uniform position initialization on $\mathcal{U}$ converges to the saddle with probability one. This is compatible with the perturbed theorem in [1], whose independent perturbation is absent here. We do not claim an open basin in the four-dimensional phase space or robustness under changes to the objective. The full time-zero phase map is singular, as the standard algorithm requires; the rank assertion concerns its given initialized surface.

The construction is necessarily nonanalytic: a globally analytic function agreeing with the indicated indefinite quadratic on an open neighborhood would agree with it everywhere, which would contradict coercivity. The example consequently makes no assertion about global analytic subclasses. Its contribution is the fixed-objective realization for the literal Schedule-Free recurrence, with the prior stable-set and smooth-construction ideas above explicitly retained.

We first analyze the two scalar quadratic recurrences. We then specify the finite data and the globally smooth function, and finish by proving that every point of (3) follows the asserted infinite trajectory.

2 The scalar quadratic recurrences

Lemma 1. *Let $A = 41/40$, $B = 9/40$, and for $k \geq 1$ define*

$$\begin{aligned} W(t) &= \left(\frac{1-t}{A-t}\right)^9, \\ X_k &= \int_0^1 t^{k-1} W(t)\, dt, \\ Z_k &= -B \int_0^1 \frac{t^{k-1} W(t)}{A-t}\, dt, \qquad Y_k = \frac{Z_k + 9X_k}{10}. \end{aligned} \tag{4}$$

Then (X_k, Z_k) solves the scalar form of (1) with gradient $-y/2$. Moreover, $X_k > 0$, $Z_k < 0$, and $Y_k > 0$; all three tend to zero, and Y_k is strictly decreasing. At every time $N \geq 1$, the span of (X_N, Z_N) is the unique line of scalar states whose negative-quadratic trajectory tends to zero.

Proof. The denominator $A - t$ is at least $1/40$ on $[0, 1]$. All integrals are finite, and

$$W'(t) = -\frac{BW(t)}{(1-t)(A-t)}, \qquad Y_k = \frac{9}{10} \int_0^1 t^{k-1} \left(\frac{1-t}{A-t}\right)^{10} dt.$$

The last integrand is positive in $(0, 1)$, so $Y_k > 0$ and $Y_{k+1} < Y_k$. The signs of X_k and Z_k are immediate, and dominated convergence gives all three limits.

Using $t = A - (A - t)$ in the integral for Z_{k+1} gives $Z_{k+1} = AZ_k + BX_k = Z_k + Y_k/4$. Integrating the derivative of $t^k(1-t)W(t)$ gives

$$(k+1)X_{k+1} = kX_k + Z_{k+1};$$

the boundary terms vanish for every $k \geq 1$. These are precisely the required updates for gradient $-y/2$.

The transition matrix in coordinates (x, z) is

$$M_k^- = \begin{pmatrix} (k+B)/(k+1) & A/(k+1) \\ B & A \end{pmatrix}, \qquad \det M_k^- = \frac{Ak}{k+1}.$$

If V_k is any other trajectory begun at time N, then

$$\det\big[(X_k, Z_k)^\mathsf{T}, V_k\big] = A^{k-N}\frac{N}{k}\det\big[(X_N, Z_N)^\mathsf{T}, V_N\big].$$

If $V_k \to 0$, the left side tends to zero. Since $A > 1$, the multiplier on the right diverges, forcing the initial determinant to vanish. Every multiple of the integral solution does tend to zero, proving the uniqueness assertion. ∎

Lemma 2. *For the scalar recurrence* (1) *with gradient* $+y/2$*, every finite initial pair gives* $x_k, z_k \to 0$*. No boundedness assumption is needed.*

Proof. The coefficients follow directly from the gradient query:

$$z_{k+1} = z_k - \frac{y_k}{4} = \frac{39}{40}z_k - \frac{9}{40}x_k, \qquad x_{k+1} = \frac{kx_k + z_{k+1}}{k+1}. \tag{5}$$

Put $a_+ = 39/40$, $d = 9/40$ and

$$E_k = dkx_k^2 + z_k^2, \qquad \mathcal{Q}_k = \frac{441x_k^2 + 79z_k^2}{1600}.$$

Expanding (5) gives the exact identity

$$E_{k+1} - E_k = -\mathcal{Q}_k + \frac{d}{k+1}\big(a_+z_k - (1+d)x_k\big)^2. \tag{6}$$

Weighted Cauchy–Schwarz yields

$$d\big(a_+z - (1+d)x\big)^2 \leq d\left(\frac{1600a_+^2}{79} + \frac{1600(1+d)^2}{441}\right)\frac{441x^2 + 79z^2}{1600} = \frac{439}{79}\mathcal{Q}.$$

Hence, for every $k \geq 5$,

$$E_{k+1} - E_k \leq -\frac{35}{474}\mathcal{Q}_k. \tag{7}$$

Every finite prefix is well-defined by (5). Starting the sum at time 5, where $E_k \geq 0$, proves $\sum_{k\geq 5}\mathcal{Q}_k < \infty$. Both coordinate square sums are finite, so both coordinates tend to zero. The same argument works from any prescribed finite starting index after its finite prefix. In particular it includes the actual standard initialization and its time-zero update. ∎

3 The finite data

We now specify all constants needed to connect an interval of equal starts to the line in Lemma 1. Put

$$K = \frac{Z_{10}}{X_{10}}, \qquad r = \frac{1143}{10000}, \qquad \delta = 10^{-6}, \qquad \epsilon = \frac{\delta}{4}, \qquad \eta = 10^{-8}.$$

Define nine rational numbers by

$$\begin{aligned} 10^6(g_0, g_1, g_2) &= (246679, 368785, 469186), \\ 10^6(g_3, g_4, g_5) &= (477024, 351755, 242007), \\ 10^6(g_6, g_7, g_8) &= (150502, 76669, 18818). \end{aligned} \tag{8}$$

Starting with $a_0 = b_0 = 1$, form for $0 \le k \le 8$

$$t_k = \frac{b_k + 9a_k}{10}, \qquad b_{k+1} = b_k - \frac{g_k}{2}, \qquad a_{k+1} = \frac{ka_k + b_{k+1}}{k+1}. \tag{9}$$

Set $t_9 = (b_9 + 9a_9)/10$, and define

$$a_{10} = \frac{9a_9}{10 - K}, \qquad b_{10} = \frac{9Ka_9}{10 - K}, \qquad g_9 = 2(b_9 - b_{10}), \qquad t_{10} = \frac{b_{10} + 9a_{10}}{10}. \tag{10}$$

The last step of (9), now using g_9, produces exactly these two states, and $b_{10} = Ka_{10}$.

To specify a common local slope, start with $P_0(h) = Q_0(h) = 1$ and define rational-coefficient polynomials for $0 \le k \le 9$ by

$$\begin{aligned} U_k(h) &= \frac{Q_k(h) + 9P_k(h)}{10}, \\ Q_{k+1}(h) &= Q_k(h) - \frac{h}{2}U_k(h), \\ P_{k+1}(h) &= \frac{kP_k(h) + Q_{k+1}(h)}{k+1}. \end{aligned} \tag{11}$$

Let h_* be the unique root in

$$H_0 = [0.5062, 0.5064]$$

of $D(h) = Q_{10}(h) - KP_{10}(h)$. Existence and uniqueness, together with all separation estimates used below, follow from the next proposition. Every decimal endpoint in its statement denotes an exact terminating rational number.

Proposition 3.1. The constants above satisfy

$$-1.272968254 < K < -1.272968253. \tag{12}$$

One has $D(0.5062) > 10^{-4}$, $D(0.5064) < -2 \cdot 10^{-4}$, and

$$-1.7 < D'(h) < -1.6 \qquad (h \in H_0).$$

Thus h_* exists uniquely in the interior of H_0. Throughout H_0,

$$|U_k(h)| < 2 \ (0 \le k \le 9), \qquad |P_{10}(h)| < 2, \quad |Q_{10}(h)| < 2, \quad P_{10}(h) > 0.15.$$

The nodes and final base state obey

$$t_i - t_{i+1} > 0.03 \ (0 \le i \le 8), \qquad t_9 - \delta > r,$$

and

$$a_{10} > 0.14, \qquad 0.1142 < t_{10} < r - \epsilon - 10^{-5}.$$

The piecewise-affine function defined below has all slopes in $(-0.999, 0.999)$ and all knot values of magnitude greater than 0.018. Its connector from $t_9 + \delta$ to $t_8 - \delta$ has positive slope greater than 0.98.

Proof. Here is an exact arithmetic certificate for these finite claims. It also specifies how the accompanying checker reproduces them without floating-point quadrature or root finding.

Expand $(A - t)^{-9}$ into its positive binomial series and integrate termwise. With $s_0 = 9!9!/19!$ and

$$\frac{s_{j+1}}{s_j} = \frac{40}{41}\frac{(j+9)(j+10)}{(j+1)(j+20)}, \tag{13}$$

one obtains

$$K = -\frac{\displaystyle\sum_{j=0}^{\infty}(j+9)s_j/41}{\displaystyle\sum_{j=0}^{\infty} s_j}.$$

Positivity justifies the interchange of sum and integral. For $M = 1501$, let S and T be the two sums truncated before M. The remaining consecutive ratios are bounded respectively by

$$q_9 = \frac{40}{41}\frac{M+9}{M+1} < 1, \qquad q_{10} = \frac{40}{41}\frac{M+10}{M+1} < 1.$$

Consequently, with

$$R_S = \frac{s_M}{1-q_9}, \qquad R_T = \frac{(M+9)s_M}{41(1-q_{10})},$$

exact rational summation gives

$$-\frac{T+R_T}{S} < K < -\frac{T}{S+R_S} \quad \text{and hence} \quad (12).$$

In particular $10 - K > 0$.

Using (11), rational interval evaluation on H_0 and (12) gives

quantity	enclosing interval
$D(0.5062)$	$(0.0001186, 0.0001188)$
$D(0.5064)$	$(-0.0002145, -0.0002143)$
$D'(H_0)$	$(-1.668, -1.662)$
$P_{10}(H_0)$	$(0.160, 0.163)$
$Q_{10}(H_0)$	$(-0.207, -0.204)$.

The asserted root bounds follow by the intermediate value theorem and strict monotonicity. The same recurrence gives the bounds on the U_k.

For clarity, the rational node values through time nine are

i	t_i	i	t_i
0	1	5	0.41635791
1	0.8766605	6	0.3239283
2	0.775244625	7	17454502/70000000
3	0.65374855	8	0.19091406875
4	0.5272137625	9	0.14656365.

Substitution in (10) encloses

a_{10}	$(0.1478, 0.1479)$
b_{10}	$(-0.1882, -0.1881)$
g_9	$(-0.0251, -0.0250)$
t_{10}	$(0.11421, 0.11423)$.

These imply the required node and final-query separations.

For completeness, let m_i denote the connector slope from $t_i + \delta$ to $t_{i-1} - \delta$. Direct interval substitution in the knot values gives

connector	slope interval
r to $t_9 - \delta$	$(0.9937, 0.9938)$
$m_i,\ 4 \le i \le 9$	$(0.9900, 0.9901)$
m_3	$(-0.0646, -0.0644)$
m_1, m_2	$(-0.9901, -0.9900)$.

The short patches have slope h_*, the central interval has slope $-1/2$, and the exterior lines have slope $1/2$. Finally, the knot ordinates specified below have magnitude greater than 0.018 by (8), the enclosure for g_9, and $|h_*\delta| < 0.507 \cdot 10^{-6}$. All denominators are positive by the node separations. Each displayed interval follows from rational operations on the given recurrences; the complete finite check is provided with the source. ∎

4 The global smooth function

Define $q_0 : \mathbb{R} \to \mathbb{R}$ by linear interpolation through the ordered knots

$$(-r, r/2), \quad (r, -r/2), \quad (t_i - \delta, g_i - h_*\delta), \quad (t_i + \delta, g_i + h_*\delta), \qquad i = 9, 8, \ldots, 0. \quad (14)$$

On the two exterior intervals set

$$q_0(t) = \begin{cases} (t+2r)/2, & t \le -r, \\ g_0 + h_*\delta + (t - t_0 - \delta)/2, & t \ge t_0 + \delta. \end{cases} \tag{15}$$

The definitions match at the outer knots. In particular, $q_0(t) = -t/2$ on $[-r, r]$, and

$$q_0(t) = g_i + h_*(t - t_i) \quad \text{on } [t_i - \delta, t_i + \delta].$$

Let ρ be the even probability density

$$\rho(s) = \begin{cases} C\exp[-1/(1-s^2)], & |s| < 1, \\ 0, & |s| \ge 1, \end{cases} \qquad \int_{-1}^{1} \rho(s)\,ds = 1.$$

It is a standard C^∞ compactly supported mollifier. Define the fixed function

$$q(t) = \int_{-1}^{1} q_0(t - \epsilon s)\rho(s)\,ds, \qquad \phi(v) = \int_0^v q(t)\,dt, \qquad f(u,v) = \frac{u^2}{4} + \phi(v). \tag{16}$$

All values in this definition have already been fixed; none depends on the initial point in (3).

Proposition 4.1. The function (16) has all the regularity, coercivity and critical-point properties in Theorem 1. Moreover,

$$\begin{aligned} q(t) &= -t/2 && (|t| \le r - \epsilon), \\ q(t) &= g_i + h_*(t - t_i) && (|t - t_i| \le \delta - \epsilon,\ 0 \le i \le 9). \end{aligned} \tag{17}$$

Proof. Proposition 3.1 makes q_0 a continuous globally 0.999-Lipschitz function with finitely many affine pieces. Convolution with ρ gives a C^∞ function, and positivity of the kernel implies

$$|q(t) - q(s)| \le 0.999|t - s|.$$

The gradient of f is $(u/2, q(v))$, so it is globally 1-Lipschitz in the Euclidean norm.

Whenever the sampling interval stays in a single affine piece, evenness of ρ and its zero first moment give $q = q_0$ exactly. This proves (17). Integration from zero then gives $f(u,v) = (u^2 - v^2)/4$ for $|v| \le r - \epsilon$. On either far tail, affine reproduction also preserves (15). Thus $\phi(v) = v^2/4 + O(|v| + 1)$ at both infinities, which proves coercivity of f.

The knot at $-r$ is positive. The knots at r and at the two ends of patch 9 are negative; every subsequent knot is positive. Together with (15), this shows that q_0 has exactly three zeros: $-2r$, 0, and the zero v_* of the connector from $t_9 + \delta$ to $t_8 - \delta$. Their slopes are $1/2$, $-1/2$ and $m_9 > 0.98$, respectively.

Smoothing preserves these zeros and creates none. The first two zeros are at distance r from their nearest knots. The third is at distance greater than $0.018/0.999 > 2\epsilon$ from

either adjacent knot, since every knot ordinate has magnitude greater than 0.018. If a point lies within ϵ of one of the three zeros, its whole sampling interval remains in the corresponding affine piece, so $q = q_0$ there. If it lies farther than ϵ from all zeros, the sampling interval has one strict sign in its interior, and its positive weighted average cannot vanish. The endpoint cases follow from the same affine reproduction or positivity argument.

Therefore $\nabla f = 0$ exactly at $(0, -2r)$, $(0, 0)$ and $(0, v_*)$. Their Hessians are

$$\operatorname{diag}(1/2, 1/2), \qquad \operatorname{diag}(1/2, -1/2), \qquad \operatorname{diag}(1/2, m_9).$$

They are all nonsingular, with the claimed saddle and minimum types. ∎

5 The initialized basin

Proof of Theorem 1. The function properties follow from Proposition 4.1. Fix an initial position $(u, 1 + e) \in \mathcal{U}$, so $|e| < \eta$. Separability of f makes its two scalar coordinate pairs independent. The first has gradient $+y/2$ and tends to zero by Lemma 2.

For the second coordinate, we prove directly from the one global function that

$$x_k^{(v)} = a_k + eP_k(h_*), \qquad z_k^{(v)} = b_k + eQ_k(h_*), \qquad 0 \le k \le 10. \tag{18}$$

The assertion holds at $k = 0$. If it holds at $k \le 9$, the actual query is $t_k + eU_k(h_*)$. Since

$$|eU_k(h_*)| < 2\eta < \delta - \epsilon,$$

formula (17) gives its exact gradient as $g_k + h_* eU_k(h_*)$. Substitution in (1), (9) and (11) proves the next step. At $k = 9$, the identities in (10) ensure that the same induction reaches the stated a_{10}, b_{10}. Thus the formula includes the real time-zero step and the final connecting step.

At time ten, both the base state and the variation satisfy the same line relation:

$$b_{10} = Ka_{10}, \qquad Q_{10}(h_*) = KP_{10}(h_*).$$

Consequently

$$z_{10}^{(v)} = Kx_{10}^{(v)}, \qquad x_{10}^{(v)} > 0.14 - 2\eta > 0.$$

The time-ten query differs from t_{10} by

$$e\frac{Q_{10}(h_*) + 9P_{10}(h_*)}{10},$$

whose magnitude is less than 2η. Proposition 3.1 therefore gives

$$0 < y_{10}^{(v)} < r - \epsilon. \tag{19}$$

Let $\lambda = x_{10}^{(v)}/X_{10} > 0$. We claim that the actual infinite tail is

$$(x_k^{(v)}, z_k^{(v)}) = \lambda(X_k, Z_k), \qquad k \geq 10. \tag{20}$$

It holds at time ten. By Lemma 1, the proposed queries λY_k are positive and decreasing, and their initial value is $y_{10}^{(v)}$. They all remain in (19), where the actual global gradient is exactly $-y/2$. The integral identities then advance the real recurrence, proving (20) inductively. Notice that only the gradient queries need stay in this interval; neither state coordinate is assumed to be there initially.

The integral tail tends to zero and remains nonzero at every finite time. Together with convergence of the positive coordinate, it proves $x_k, z_k, y_k \to 0$ for every original initial point in $\mathcal{U}$. Every finite step is well-defined, since the objective is globally smooth and $k + 1 \neq 0$. The rectangle has area $4\eta > 0$.

It remains to check rank preservation. For any $c \in (0, 1)$, set

$$G_c(w) = w - c\nabla f(w).$$

For every target b, its inverse equation is $w = b + c\nabla f(w)$, a contraction on $\mathbb{R}^2$ with constant at most c. It has a unique solution. Moreover, $DG_c = I - c\nabla^2 f$ is invertible everywhere, so the inverse function theorem makes this global inverse C^∞.

Given the output (x', z') of a phase update with $k \geq 1$, recover successively

$$x = \frac{(k+1)x' - z'}{k}, \qquad y = G_{1/20}^{-1}\left(\frac{z' + 9x}{10}\right), \qquad z = 10y - 9x. \tag{21}$$

Indeed the gradient update is $z' = 10y - 9x - \nabla f(y)/2$. Thus (21) is an everywhere-defined smooth inverse of the actual phase map. At time zero, standard initialization gives $(x_1, z_1) = (G_{1/2}(a), G_{1/2}(a))$, a rank-two embedding because $G_{1/2}$ is a global diffeomorphism. Composing with all subsequent phase diffeomorphisms preserves that rank at every finite time. This completes the proof. ∎

The constructed basin is an initialized-surface phenomenon. It realizes a global intersection that local diagonal transversality and ambient nullity alone do not determine. The independent perturbation in [1] changes that initialized distribution, so its theorem remains applicable to the perturbed method on this same objective. The example does not identify one-time noise as the unique remedy, and it makes no claim about the frequency of this behavior for practical objectives.

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